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Department of Computer Science and Engineering

NAAC A

# NEURO DIAGNOSIS AI

*AI-Powered Brain Tumor Classification & Explainability System*

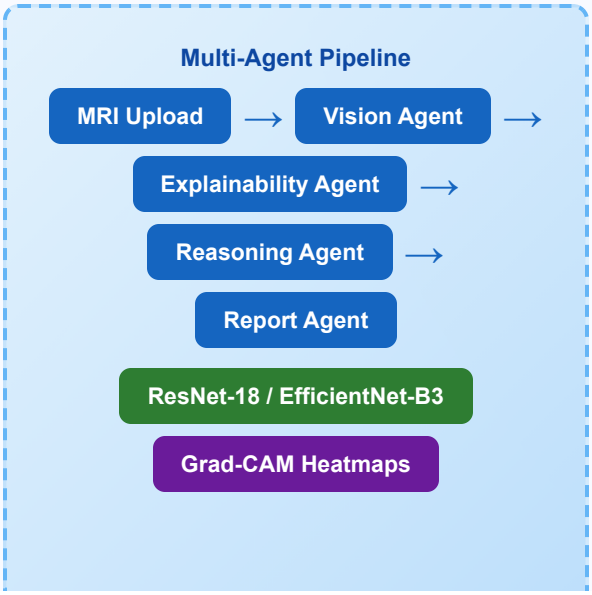
## ABSTRACT

Brain tumors are among the most critical neurological conditions, requiring accurate and timely diagnosis for effective treatment planning. Magnetic Resonance Imaging (MRI) is the gold standard for brain tumor detection, yet manual interpretation by radiologists is time-consuming, subjective, and prone to inter-observer variability. This project presents **Neuro Diagnosis AI**, an AI-powered web-based system that automates brain MRI classification into four categories — *Glioma, Meningioma, Pituitary Tumor, and No Tumor* — using deep learning models (ResNet-18 and EfficientNet-B3). The system integrates **Grad-CAM** (Gradient-weighted Class Activation Mapping) for visual explainability, highlighting the regions of the MRI that influenced the model's prediction. Additionally, a **Gemini AI-powered Reasoning Agent** generates concise medical explanations, providing clinical context and recommended next steps. The multi-agent architecture — comprising Vision, Explainability, Reasoning, and Report agents — ensures a comprehensive, transparent, and clinically meaningful diagnostic workflow. Built with a FastAPI backend and React frontend, the system is cost-effective, scalable, and designed to assist healthcare professionals in early detection and improved patient outcomes.

## DESCRIPTION

Modern healthcare increasingly relies on AI-assisted diagnostic tools, yet brain tumor assessment remains heavily dependent on manual MRI interpretation, which is subjective and resource-intensive. Traditional diagnostic approaches require expert neuroradiologists and sophisticated imaging techniques, making the process complex and expensive. Early detection is often delayed due to the lack of automated solutions, leading to worsening conditions for patients.

## SYSTEM ARCHITECTURE



To overcome these challenges, we developed **Neuro Diagnosis AI**, an intelligent multi-agent system that automates brain tumor classification using deep learning. The system classifies brain MRI scans with high accuracy, reduces dependency on manual evaluation, and provides visual explainability through Grad-CAM overlays. Our system is accessible, efficient, and scalable, making advanced neurodiagnostics available to a wider range of healthcare providers.

Gemini AI Explanations

## EXISTING SOLUTIONS

1. Manual MRI interpretation by radiologists
2. Basic Convolutional Neural Networks (CNNs)
3. Traditional Machine Learning with hand-crafted features
4. Transfer Learning models without explainability
5. Single-model classification without clinical context

## TECHNOLOGY STACK

PyTorch | ResNet-18 | EfficientNet-B3 | FastAPI | React.js | Grad-CAM | Gemini AI | Vite | Docker | Python | JavaScript

## CONCLUSION

This project aims to automate and enhance brain tumor diagnosis using deep learning and AI-driven explainability. Currently, manual MRI interpretation is time-consuming and subjective, leading to inconsistencies in diagnosis. Traditional methods rely on expert neuroradiologists, making large-scale screening challenging. In our system, Neuro Diagnosis AI, we leverage pre-trained deep learning architectures (ResNet-18 and EfficientNet-B3) to classify brain MRI scans into four categories accurately. The integration of Grad-CAM provides visual transparency, while the Gemini-powered Reasoning Agent delivers clinically meaningful explanations. By combining deep learning, visual explainability, and natural language reasoning in a multi-agent architecture, the system provides quick, reliable, and interpretable brain tumor assessment from MRI images. This solution is efficient, scalable, and easy to deploy, ensuring early detection and improved patient care.

## TEAM DETAILS

**Batch - A6**

## PROJECT GUIDE

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